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NON-SHOCK INITIATION MODEL FOR EXPLOSIVE FAMILIES: EXPERIMENTAL RESULTS

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Abstract. The "DaMaGe-Initiated-Reaction" (DMGIR) computational model has been developed to predict the response of high explosives to non-shock mechanical insults. The distinguishing feature of this model is the introduction of a damage variable, which relates the evolution of damage to the initiation of a reaction in the explosive, and its growth to detonation. Specifically designed experiments were used to study the initiation process of each explosive family with embedded shock sensors and optical diagnostics. The experimental portion of this model development began with a study of PBXN-5 to develop DMGIR model coefficients for the rigid plastic bonded family, followed by studies of the cast, and bulk-moldable explosive families. The experimental results show an initiation mechanism that is related to input energy and material damage, with well defined initiation thresholds for each explosive family. These initiation details will extend the predictive capability of the DMGIR model from the rigid family into the cast and bulk-moldable families.

Keywords: initiation, detonation, deflagration, damage, relief wave, rarefaction, porosity

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INTRODUCTION

The impulsive load caused by the impact of a projectile or fragment into an energetic material will produce a range of responses that can span from mechanical damage with no reaction to a self-sustaining deflagration that consumes the material at an increasing rate, transitioning into a detonation.

The magnitude of impulsive energy input, and the extent to which damage is generated in the energetic material have a direct effect on the onset of, and the extent of reaction produced in an explosive sample. Impact-induced deflagrations with sufficient energy input and mechanical damage have been observed to transition within a few microseconds into a growth to detonation. The generation of damage is dependent on the shock loading and unloading profile, and the material strength. Numerical simulations of this damage are

used to relate shear-induced strain to explosive initiation by the use of a damage variable which is embedded in the DMGIR non-shock initiation model. This treatment of non-shock initiation allows all explosives to be described numerically in one of the following three explosive families; rigid plastic bonded, cast, and moldable plastic explosives.[1]

An experimental program was designed which made use of embedded shock sensors and optical diagnostics in order to better understand the explosive initiation process of these explosive families from impulsive, non-shock loading.

EXPERIMENTAL DESCRIPTION

A single-stage light gas, or a propellant gun was used in each test to accelerate a projectile to the desired velocity, impacting an explosive test specimen. Projectile free flight of ~8-feet allowed

blast protection for the gun. The test specimen was either a solid or a sandwiched cylindrical billet, instrumented with PVDF piezoelectric shock sensors for velocity measurements.[2] Digital framing cameras recorded the details of projectile impact and subsequent reaction on each experiment. The test configuration is shown schematically in Figure 1.

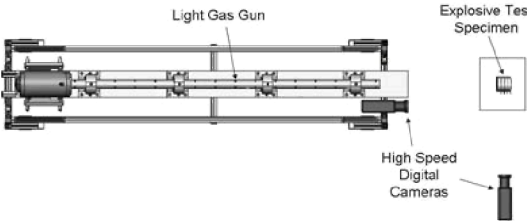


Figure 1. Gun configuration for explosive impact study.

The projectile geometry consisted of a blunt-faced, hollow, right circular cylinder with a 0.500-inch diameter, with O-ring groove near the impact face. Projectile materials were aluminum or brass.

This projectile design provided a planar, short duration shock loading pulse, followed immediately by axial and radial rarefaction waves which provide well controlled mechanical damage, as shown in Fig. 2. This controlled damage was intended to simulate the expected profile produced by accidental impact.[3] Impact velocity capabilities range from 0.20 – 1.66 km/s. Explosively driven water blades were also used to generate mechanical damage, although less well controlled.

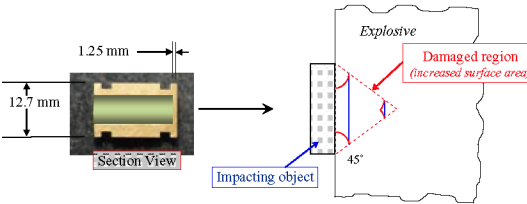


Figure 2. Shock and release from projectile impact generates controlled damage region in explosive.

TEST SPECIMEN

Explosives are composite materials with explosive crystals embedded in a variety of polymer-based binders, with mechanical properties

that depend mainly on the mixture used to form the polymer binder. The three explosive families described in this study were represented by PBXN-5, ρ 1.60 gm/cc, (rigid plastic bonded), cast composition-B, ρ 1.61 gm/cc, (cast), and PETN- or RDX-based Primasheet 1000, 2000, and composition -4, ρ 1.60 gm/cc (moldable plastic).

The test specimen configurations consisted of a cylindrical billet that was made from sandwiched billets with piezoelectric shock sensors embedded at each interface. Billet diameters were nominally 4.0–inch diameter. The piezoelectric shock sensors measured stress-time history along the axial centerline of the PBXN-5 billet. Shock front arrival times for each sensor location are plotted in Fig. 3 for a collection of PBXN-5 tests, showing the transition region between mechanical response and deflagration growth to detonation, as well as prompt detonation response from detonator input.

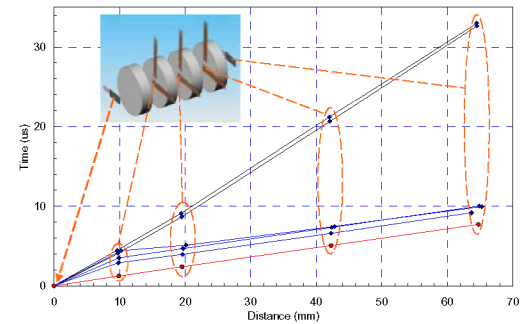


Figure 3. Sandwiched billet of PBXN-5 with embedded piezoelectric shock sensors, and shock arrival data.

The embedded Bauer PVDF shock sensor thickness was 25 μ m, insulated on both sides with 50 μ m Teflon . MSI Piezofilm sensors were used on front and rear surfaces for timing measurements, augmenting the embedded shock sensor measurements on the sandwiched billets. The impact surface Piezofilm sensor also provided the instrumentation trigger signal.

SHOCK PROPAGATION RESULTS

The piezoelectric output from each shock sensor provides a wave propagation measurement

with a timing uncertainty of ± 4 ns, based on the 2 ns sampling interval. Average velocity between each shock sensor location is based on thickness measurements along the billet centerline and transit time between sensor locations. Shock propagation time through each sensor package is removed prior to velocity calculations. Prompt-detonation control experiments were conducted to measure material response for each family under Shock-to-Detonation Transition (SDT) conditions. The PBXN-5 data set from Fig. 3 is plotted in velocity-vs-distance space in Fig. 4 to illustrate the observed non-shock initiation behavior for PBXN-5.

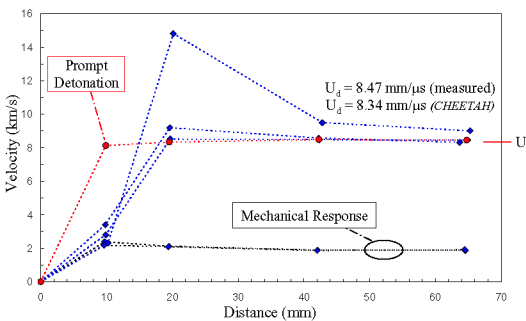


Figure 4. Velocity calculations from PBXN-5 tests show non-shock initiation results, prompt detonation results, and CHEETAH predictions.

The non-shock insults that result in a mechanical response are observed to initially propagate above the PBXN-5 acoustic velocity (1.35 km/s), with a slight decrease in velocity as a function of distance. [4] The prompt-detonation control experiments show a velocity measurement that asymptotically approaches a steady velocity of 8.47 km/s, which shows that the CHEETAH calculations estimate of 8.34 km/s, is within 1.5% of the experimental velocity. The non-shock, or damage initiated tests show the initial velocity to be above the detonation velocity followed by a rapid increase through the next 10-mm thick disc to approximately the detonation velocity. The observed transient overshoot in velocity is consistent with a Deflagration-to-Detonation Transition (DDT) in porous explosives, in which the porosity is caused by mechanical damage. This collection of test data at increasing impact

velocities show that the non-shock initiation behavior, as shown in Figures 3 and 4, is dependent on input energy.

The average shock velocity through the assembled explosive billet is dependent on the specific energy input, as shown in Fig. 5 for rigid and cast materials, plotted on a semi-log scale.

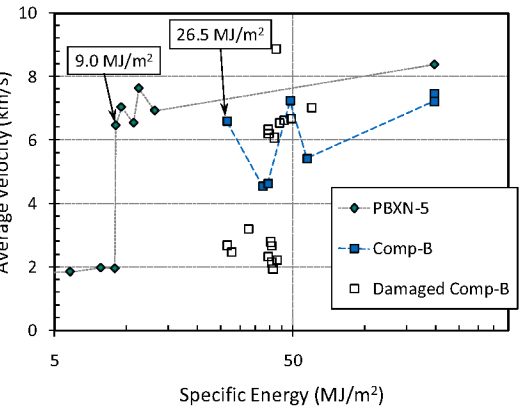


Figure 5. Average velocity through explosive billet transitions to near-detonation velocity at initiation threshold. Damaged Comp-B has higher initiation threshold with more scatter than undamaged Comp-B.

The average shock velocities range from slightly above acoustic velocity for mechanical response to nearly detonation velocity for a SDT-type reaction, with the velocity for non-shock initiations ranging between these limits. For energy inputs above the initiation threshold, the measured shock velocity is dependent on the run distance to the DDT transition. Baseline tests used a M16 non-electric detonator for a SDT initiation, with a specific energy of 195.3 MJ/m².

All damage initiation tests were focused below the SDT threshold energy level. PBXN-5 response showed a well defined initiation threshold of 9.0 MJ/m², with modest scatter in the average velocity above the initiation threshold. The initiation threshold has not been identified for Comp-B, but tests above 26.5 MJ/m² show a similar initiation response to PBXN-5, with test results from damaged Comp-B showing an initiation threshold above the undamaged Comp-B. The scatter in average velocity-vs-input energy results appears reasonable given that the extent of damage consists of cracks and voids with a length scale of several

millimeters. Damaged Comp-B appears to be less sensitive to non-shock initiation than undamaged Comp-B.

The moldable explosive family was investigated initially with a sandwiched billet configuration made with multiple layers of C2 Primasheet 1000 (PETN), and 2000 (RDX) stacked and clamped together. Subsequent experiments with bulk moldable Composition C-4 used low density confinement rings to ensure a density of $\rho=1.60$ gm/cc. The results for the moldable family are shown in Fig. 6.

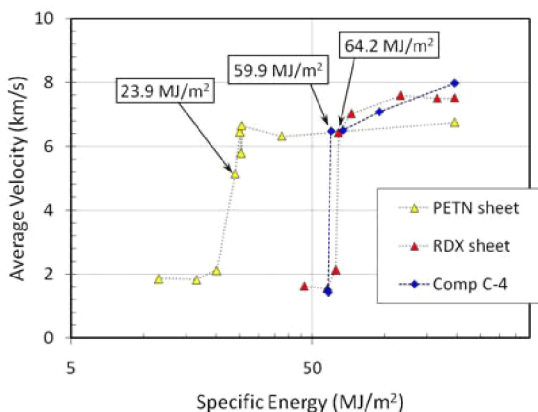


Figure 6. Initiation thresholds for the moldable plastic explosive family, using both sheet and bulk material.

The initiation response of the moldable explosive family demonstrates well defined thresholds for the two RDX-based materials, and a more gradual threshold for the PETN material.

SUMMARY

The initiation mechanism for these projectile-impact experiments appears to be related to input energy and material damage, which is caused by shock and release wave interactions at the impact site that create additional surface area. The impulsive loading has sufficient time duration to compact this damaged material, initiating a DDT reaction, as evidenced by the transient super detonation velocities. The initiation thresholds are well defined for each explosive family. The cast Comp-B material has cracks and voids with a

length scale of several millimeters, and this condition appears to have a higher initiation threshold, with more scatter than the undamaged Comp-B. The RDX sheet tests show that projectile impact at energy levels above 100 MJ/m^2 result in average velocities similar to the SDT initiation tests, even though the material experiences extensive damage from the shock loading and release profiles.

This study is directly coupled to the Damage-Initiated Reaction (DMGIR) model development, which has demonstrated the predictive capability to capture these effects. [5]

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